

Lab program-6:

6. Construct a C program to simulate Round Robin scheduling algorithm with C.

```
C lab_6.c > main()
1  #include<stdio.h>
2  #include<conio.h>
3  int main()
4  {
5  int i, NOP, sum=0, count=0, y, quant, wt=0, tat=0, at[10], bt[10], temp[10]; float
6  avg_wt, avg_tat;
7  printf(" Total number of process in the system: ");
8  scanf("%d", &NOP);
9  y = NOP;
10 for(i=0; i<NOP; i++)
11 {
12 printf("\n Enter the Arrival and Burst time of the Process[%d]\n", i+1);
13 printf(" Arrival time is: \t");
14 scanf("%d", &at[i]);
15 printf(" \nBurst time is: \t");
16 scanf("%d", &bt[i]); temp[i] =
17 bt[i];
18 }
19 printf("Enter the Time Quantum for the process: \t");
20 scanf("%d", &quant);
21 printf("\n Process No \t\t Burst Time \t\t TAT \t\t Waiting Time ");
22 for(sum=0, i = 0; y!=0; )
23 {
24 if(temp[i] <= quant && temp[i] > 0)
25 {
26 sum = sum + temp[i];
27 temp[i] = 0;
28 count=1;
29 }
30 else if(temp[i] > 0)
31 {
32 temp[i] = temp[i] - quant;
33 sum = sum + quant;
34 }
35
36 if(temp[i]==0 && count==1)
37 {
38 y--;
39 printf("\nProcess No[%d] \t\t %d\t\t\t %d\t\t\t %d", i+1, bt[i], sum- at[i], sum- at[i]-bt[i]);
40 wt = wt+sum-at[i]-bt[i];
41 tat = tat+sum-at[i]; count =0;
42 }
43 if(i==NOP-1)
44 {
45 i = 0;
46 }
47
48 else if(at[i+1]<=sum)
49 {
50 i++;
51 }
52
53 else
54 {
55 i = 0;
56 }
57 }
58 avg_wt = wt * 1.0/NOP;
59 avg_tat = tat * 1.0/NOP;
60 printf("\n Average Turn Around Time: \t%f", avg_wt);
61 printf("\n Average Waiting Time: \t%f", avg_tat);
62 getch();
63 }
```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_6.exe
Total number of process in the system: 3

Enter the Arrival and Burst time of the Process[1]
Arrival time is:      4

Burst time is: 45

Enter the Arrival and Burst time of the Process[2]
Arrival time is:      6

Burst time is: 123

Enter the Arrival and Burst time of the Process[3]
Arrival time is:      4

Burst time is: 6
Enter the Time Quantum for the process:      3

Process No      Burst Time      TAT      Waiting Time
Process No[3]    6      17      11
Process No[1]    45     86     41
Process No[2]    123    168     45
Average Turn Around Time:      32.333332
Average Waiting Time: 90.333336
```